

Food Delivery Booking

Problem Statement

A food delivery company has '**n**' number of delivery executives. For simplicity, take the count as 5 but the program should work for any number of delivery executives (Let their names be identified as DE1, DE2,DE-n).

There are only 5 restaurants in the city for pickup and 5 drop locations (Each location can have multiple customers). After delivering a food package, the delivery executive waits there for delivery allotment.

Each customer is identified uniquely by a **Customer-ID**.

Constraints

1. Delivery charge for every single order is **Rs 50** for the delivery executive.
 2. If multiple orders (say **n**) are from the same delivery location within **15 mins** period, combine orders to a maximum **5 per delivery executive**.
In such case, the delivery charge will be base rate: **Rs.50 + Rs.5 for every other order**
— Formula: $(50 + 5 \times (n-1))$.
 3. An allowance of **Rs.10** will be given for every trip made. Combined orders will be counted as a single trip.
 4. Assign the subsequent bookings giving preference to the executive who has earned the **least delivery charge** among the other available delivery executives (excluding trip allowance).
 5. Every trip will take **30 mins** to reach the destination.
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Questions

- Q1. Write a function to **handle booking**.
 - Q2. Write a function to **assign delivery executive**.
 - Q3. Write a function that can **display delivery executive's activity thus far**. This should contain commission earned, allowance earned (calculated based on criteria 2 and 3).
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Sample Inputs & Expected Outputs

Input 1

Customer ID	1
Restaurant	A
Destination Point	D
Time	9:00 AM

Expected Output:

Output Booking ID	1
Available Executives	All (DE1 to DE5)
Delivery Charge Earned	DE1: 0, DE2: 0, DE3: 0, DE4: 0, DE5: 0
Allotted Delivery Executive	DE1

Input 2

Customer ID	2
Restaurant	B
Destination Point	A
Time	10:00 AM

Expected Output:

Output Booking ID	2
Available Executives	All (DE1 to DE5)
Delivery Charge Earned	DE1: 50, DE2: 0, DE3: 0, DE4: 0, DE5: 0
Allotted Delivery Executive	DE2

Input 3

Customer ID	3
Restaurant	B
Destination Point	A
Time	10:10 AM

Expected Output:

Note: Customer 3 orders from Restaurant B to Destination A at 10:10 AM. Customer 2 had the same destination (A) at 10:00 AM and was assigned DE2. Since both are within 15 mins and same destination, orders get combined with DE2.

Output Booking ID	3
Available Executives	All (DE1 to DE5)

Delivery Charge Earned	DE1: 50, DE2: 50, DE3: 0, DE4: 0, DE5: 0
Allotted Delivery Executive	DE2 (combined with Booking 2)